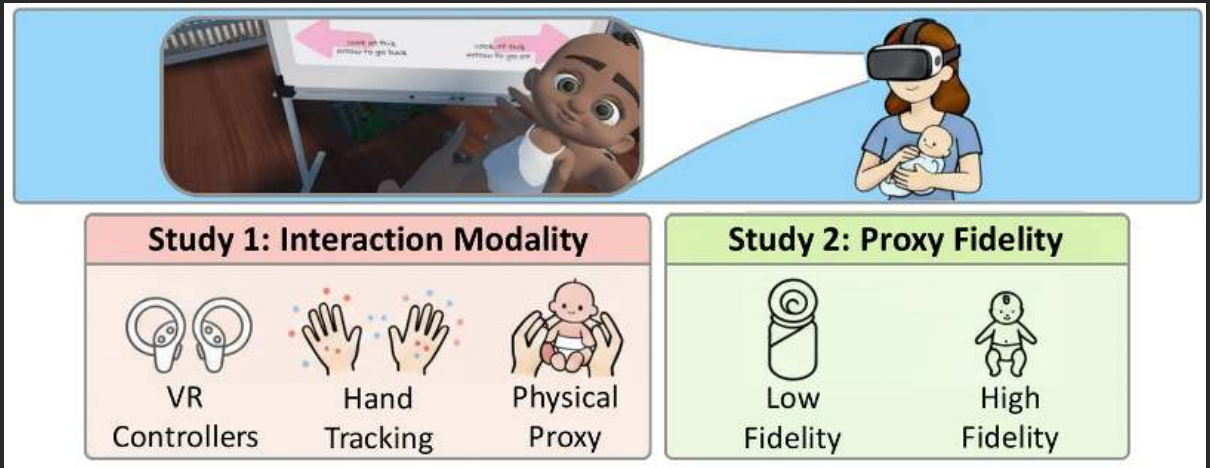
Master's Degree in Cinema & Media Engineering

Exploring Virtual Reality for Breastfeeding Education

A Study on Interaction Methods & Proxy Fidelity



Supervisors:
Prof. Fabrizio Lamberti, Prof. Alvaro Uribe Quevedo,
Eng. Alessandro Visconti, Prof. Bill Kapralos

Candidate:
Virginia Ermondi

ERASMUS +
Non-EU Partner Countries

Final Project



OntarioTech UNIVERSITY

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Context

Breastfeeding & Latching Education



Latching

Procedure through which the caregiver reads the infant behavioral cues and reacts by bringing the infant's nose towards the breast to initiate a breastfeeding session.

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02

Research Questions

Interaction Methods & Proxy Fidelity on Usability, Task Load and Presence

Interaction Methods

How a user interacts with VR

Proxy

Stand-in object used to represent something else during a simulation

Proxy Fidelity

Degree to which a proxy accurately represents its real-world counterpart

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01

Context

Breastfeeding & Breastfeeding Education

48%

of infants under six months old is exclusively breastfed

World Health Organization

48%

of infants under six months old is exclusively breastfed



World Health Organization

Latching Education

Current Situation

01 Traditional

Prenatal classes, consultations, ...

Useful

Structural challenges
Availability of material

02 ICT-based

Mobile apps, SMS, telehealth programs, ...

On demand

Lack of immersion
Lack of hands-on practice

03 XR/VR

Few applications exist

Immersion
Hands-on practice

Little explored

02

Research Questions

Interaction Methods & Proxy Fidelity on Usability, Task Load and Presence

Usability, Task Load, Presence

Usability

How easy and efficient it is for users to interact with and navigate a VR system



Task Load

Amount of mental, physical, temporal effort required to understand and perform tasks in VR



Presence

Feeling of being physically and psychologically “inside” the virtual environment



Research Questions (RQs)

01

How does the interaction method affect usability, task load and presence in VR breastfeeding education?



02

When a proxy is used, how does its fidelity affect usability, task load and presence in VR breastfeeding education?



RQ 01: Interaction Methods

Controllers

Precision

Lack of naturalness

Widely used



Hand Tracking

Naturalness

Reduced precision

Emerging method



Proxy

Realism

Mismatch

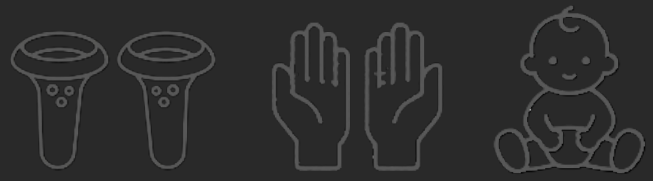
Little exploration



Research Questions (RQs)

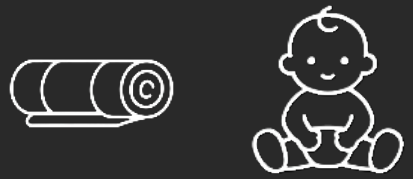
01

How does the interaction method affect usability, task load and presence in VR breastfeeding education?



02

When a proxy is used, how does its fidelity affect usability, task load and presence in VR breastfeeding education?



RQ 02: Proxy Fidelity





Low-Fidelity

- Easy to build ✓
- Less realistic ✗

High-Fidelity

- More realistic ✓
- Harder to get ✗

03

Implementation

Creating the Breastfeeding VR Scenario

VR Breastfeeding Scenario

01

Tutorial

- User embodies a caregiver
- Environment
- Holding
- Alignment
- Latching



VR Breastfeeding Scenario

02

Serious Game

- 5 attempts
- 4 seconds
- Performance
- Immediate feedback



VR Breastfeeding Scenario



Final score

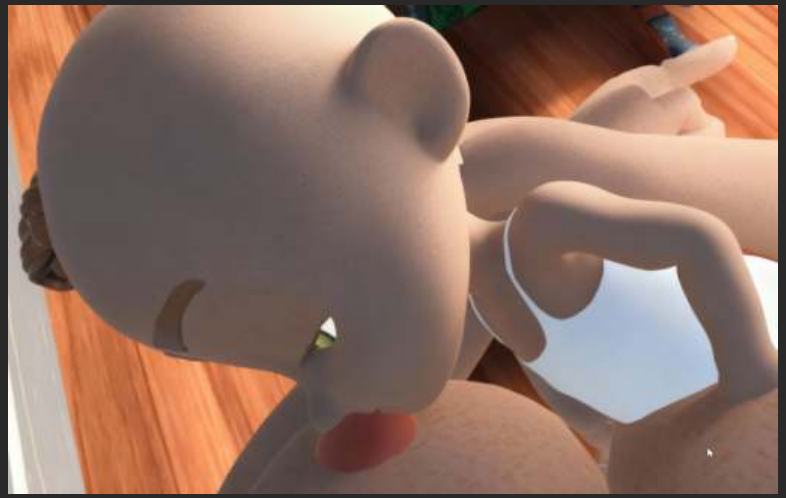


Whiteboard + Eye gaze

VR Breastfeeding Scenario



Cartoon-style aesthetic



Body tracking

04

Experiments

Exploratory Within-Subjects Studies

Study 01 - Interaction Methods

- Participants: women
- Sample size n = 4
- Latin square design
- 3 conditions







Study 02 - Proxy Fidelity

- Participants: women
- Sample size n = 8
- Randomization
- 2 conditions






Methods of Evaluation

Usability, Task Load, Presence, Open-ended Feedback

Usability

System Usability Scale (SUS)

Effectiveness, efficiency, satisfaction

Load

Raw Task Load Index (R-TLX)

Mental, physical, temporal demand, performance, effort, frustration

Presence

Presence Questionnaire (PQ)

Involvement, engagement

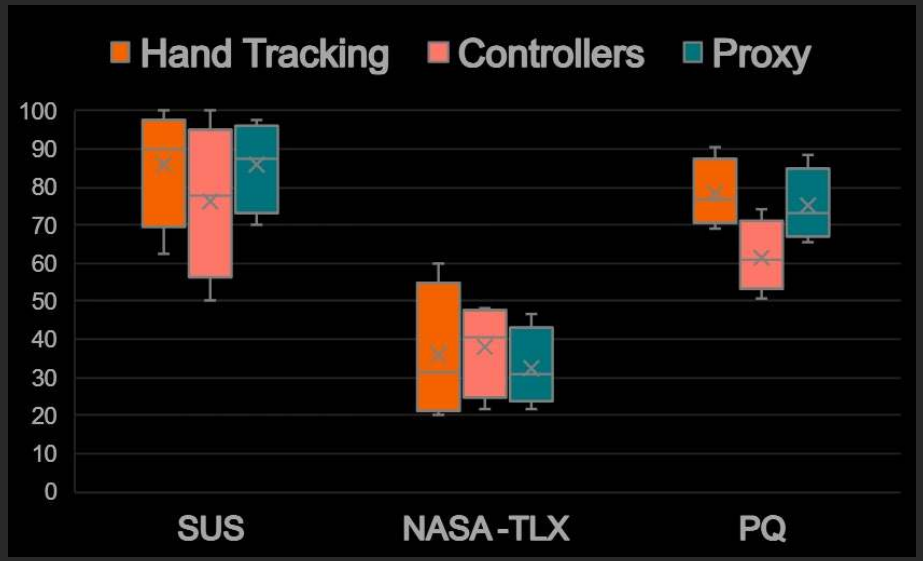
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Results

Usability, Task Load, Presence

Study 01 - Results

Usability, Task Load, Presence Questionnaire



Usability (SUS)	
Controllers	76.3, good
Hand-tracking	85.6, excellent
Proxy	85.6, excellent

Presence (PQ)	
Controllers	61.7, moderate
Hand Tracking	81.3, good
Proxy	75, good

Task Load (R-TLX)	
Controllers	37.9, moderate
Hand-tracking	35.8, low
Proxy	32.5, low

Study 01 - Results

Open-ended Feedback

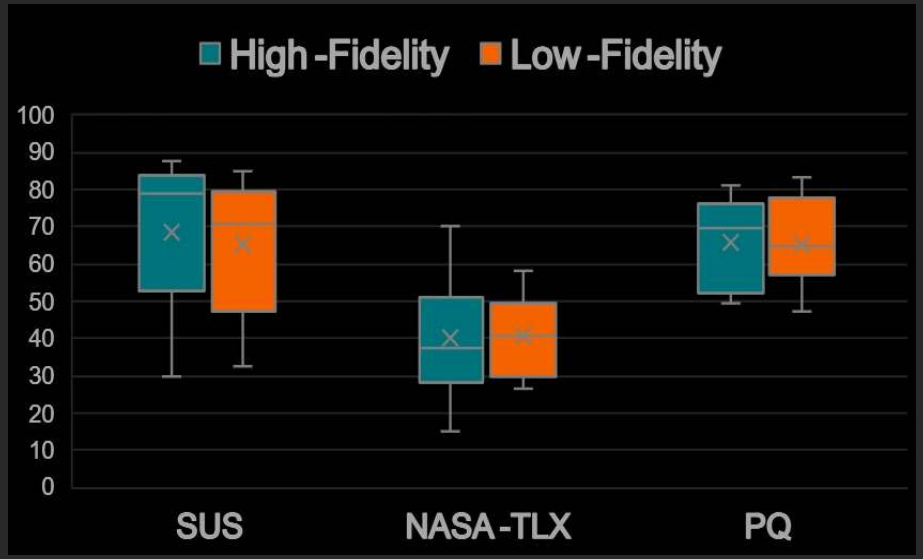
"The controller modality of interaction was not involving."

"The hand-tracking and proxy modalities were extremely immersive."

"The proxy modality of interaction felt really realistic, as I was actually holding an infant."

Study 02 - Results

Usability, Task Load, Presence Questionnaire



Usability (SUS)	
High-fidelity	68.8, good
Low-fidelity	68.8, good

Presence (PQ)	
High-fidelity	65.9, good
Low-fidelity	65.4, good

Task Load (R-TLX)	
High-fidelity	40.0, moderate
Low-fidelity	40.6, moderate

Study 02 - Results

Open-ended Feedback

“The mere presence of an object was enough to make the experience believable.”

“I did not care about the realism of the proxy.”

06

Conclusions

Thesis Work Results

01 RQ 01

How does the interaction method affect usability, task load and presence in VR Breastfeeding Education?

It is possible that interaction method affects usability, task load and presence

02 RQ 02

When a proxy is used, how does its fidelity affect usability, task load and presence in VR Breastfeeding Education?

It is possible that the proxy fidelity does not affect usability, task load and presence

Practical Implications

01 Natural methods of interaction might be the most suitable ones for VR breastfeeding education

02 If proxies are used, low -fidelity ones may be used without compromising the experience

Future Work

01 Expand the studies and increase the sample sizes to have statistically more significant results

02 Allow for customization of the avatars (both the caregiver and the infant)

03 Further study the minimal needed fidelity of the proxy to explore how “low in fidelity” it is possible to go

Real-World Scenarios

01 Prenatal classes

02 Telehealth education

03 Medical training simulations

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Q&As

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